Rafael Antonio Tanda Pretel

Software Engineer — Backend & offline-first systems

Havana, Cuba · relocating to La Palma, Spain · Spanish citizen · EU work authorization
tandapretelrafael2005@gmail.com · github.com/Ranta2025 · rafael-tanda.vercel.app

PROFILE

Software engineer who builds systems that keep working on low-end hardware and unreliable networks. I design backends in Go and TypeScript and offline-first clients in Flutter, Godot and Python, with local data as the source of truth, explicit sync, hard performance budgets and tests against real databases. Every project below is documented and tested.

SELECTED PROJECTS

ChatApp — real-time messaging platform Jan 2026 – present

`Go, Gin, React 19, PostgreSQL, Redis, MinIO, WebSockets, Docker, Playwright · github.com/Ranta2025/chatapp`

- Designed a 7-service platform (API, Postgres, Redis, MinIO, Nginx, frontend, Cloudflare Tunnel) with 1:1 and group chat, delivery receipts, presence, media, voice notes, video calls and Web Push.
- Hardened auth with 15-min JWTs in HttpOnly cookies, rotated revocable refresh tokens, bcrypt, account lockout and per-IP rate limits; group permissions enforced server-side on REST and WebSocket.
- Built a PWA with an IndexedDB offline outbox that delivers queued messages in order on reconnect; 276 test files across Go and frontend, plus Playwright e2e in GitHub Actions CI.

OrtoSim CUBA — offline medical training game (undergraduate thesis) Oct 2026 – present

`Godot 4, GDScript, GUT, PocketBase, Caddy, JSON Schema, Blender, Docker`

- Architected a serious game for a medical master's program targeting 2 GB RAM Android phones: < 60 MB APK, 30 FPS, < 100 draw calls, open-source tooling only due to sanctions.
- Applied hexagonal architecture in Godot: a pure domain behind five ports, unit-tested with GUT (150 test files) using injected clock and randomness.
- Made the case engine data-driven (JSON Schema-validated cases, 12 step kinds) and shipped content as signed .pck packs with a SHA-256 manifest and offline event queue synced via PocketBase.

Salary Fund Distribution — Acuerdo 10199/2025 Oct 2026

`Python, tkinter, pytest, PyInstaller · github.com/Ranta2025/fondo-salario-acuerdo-10199`

- Built an offline desktop app that applies a 2025 Cuban regulation to distribute unexecuted salary funds, with configurable weights so each unit can match its own rules.
- Guaranteed totals reconcile to the cent with largest-remainder rounding; isolated domain logic covered by 100+ pytest tests; auditable Excel export.

Salon Manager — offline-first mobile app for a real salon Sep 2026

`Flutter, Dart, sqflite, Provider · github.com/Ranta2025/gestion-salon`

- Delivered finances, clients, appointments with reminders, cash closing and PDF/CSV reports running fully on-device with no backend.
- Tested repositories against real in-memory SQLite and made restore validate backups before touching any data.

XILEF — offline licensing for a commercial product Oct 2026 – present

`TypeScript, Node.js, Ed25519`

- Ed25519 license issuance with monotonic sequence numbers for renewal and revocation; every artifact is self-verified before it is written.

SKILLS

Languages	Go, TypeScript, Python, Dart, GDScript, Java
Backend & data	Gin, REST, WebSockets, PostgreSQL, Redis, SQLite, PocketBase, S3/MinIO
Clients	React 19, PWA (service workers, IndexedDB, Web Push), Flutter, Godot 4
Quality & delivery	go test, Playwright, pytest, GUT, GitHub Actions, Docker Compose, Nginx, Caddy

EDUCATION

B.Eng. in Computer Science (Ingeniería en Ciencias Informáticas), University of Information Sciences (UCI), Havana — 3rd of 4 years, expected 2028

LANGUAGES

Spanish — native · English — B1 (certification exam March 2027)